

If already registered, click to check your payment status

Course outline
About NPTEL
How does an NPTEL online course work?
Week 1
Week 2
Week 3
week 4
Week 5
Week 6
Week 7
Week 8
Tuples- Python Data Structure Lottery Simulation - Profit or Loss Lottery Simulation - Profit or Loss - Part 01 Lottery Simulation - Profit or Loss - Part 02 Lottery Simulation - Profit or Loss - Part 03 Lottery Simulation - Profit or Loss - Part 04 Lottery Simulation - Profit or Loss - Part 05 Lottery Simulation - Profit or Loss - Part 06 Image Processing - Enhance your Images Image Processing - Enhance your Images - Part 01 Image Processing - Enhance your Images - Part 02 Image Processing - Enhance your Images - Part 03
Anagrams Anagrams - Part 01 Anagrams - Part 02 Anagrams - Part 03 Facebook Sentiment Analysis - Part 01 Facebook Sentiment Analysis - Part 02 Facebook Sentiment Analysis - Part 03 Facebook Sentiment Analysis - Part 04 Week 8 Feedback Form: The Joy of Computing using Python
Quiz: Week 8 : Assignment 8 Week 8: Programming Assignment 1 Week 8: Programming Assignment 2 Week 8: Programming Assignment 3 Week 8 Solution
Week 9
Week 10
Week 11
Week 12
Text Transcripts
Download Videos
Problem Solving Session - July 2025

Week 8 : Assignment 8

The due date for submitting this assignment has passed.

Due on 2025-09-17, 23:59 IST.

Assignment submitted on 2025-09-17, 17:54 IST

An anagram is a word formed by rearranging the letters of another word. For example, the word "listen" can be turned into "silent". Both words use the exact same letters, just in a different order.

Anagrams are often used in word puzzles and games. Some fun examples include:

- evil → vile
- dusty → study
- save → vase

To check if two words are anagrams, you can sort their letters and see if they match.

- 1) Which of the following is an example of an anagram?

- ☐ cat → dog
☒ loop → pool
☐ red → reed
☐ cup → mug

Yes, the answer is correct.
Score: 1
Accepted Answers: loop → pool

- 2) What will be the output of the following code?

```
def are_anagrams(w1, w2):
    return sorted(w1) == sorted(w2)

print(are_anagrams("listen", "silent"))
```

- ☒ True
☐ False
☐ Error: cannot compare lists
☐ It depends on the length of the words

No, the answer is incorrect.
Score: 0
Accepted Answers: False

- 3) What will be the output of the following code?

```
def are_anagrams(w1, w2):
    w1 = w1.replace(" ", "").lower()
    w2 = w2.replace(" ", "").lower()
    return sorted(w1) == sorted(w2)

print(are_anagrams("Dormitory", "Dirty room"))
```

- ☒ True
☐ False
☐ False because the strings are of different original lengths
☐ Depends on whether the input has punctuation

Yes, the answer is correct.
Score: 1
Accepted Answers: True

- 4) How many distinct anagrams can be formed using all the letters of the word "LEVEL"?

- ☐ 60
☐ 120
☒ 30
☐ 20

Yes, the answer is correct.
Score: 1
Accepted Answers: 30

- 5) What will be the output of the following code?

```
words = ["listen", "silent", "enlist", "google", "gogole"]

anagrams = []
for i in range(len(words)):
    for j in range(1 + 1, len(words)):
        if sorted(words[i]) == sorted(words[j]):
            anagrams.append((words[i], words[j]))

print(anagrams)
```

- ☐ [('listen', 'silent'), ('listen', 'enlist'), ('silent', 'enlist'), ('google', 'gogole')]
☒ [('listen', 'silent'), ('silent', 'enlist'), ('google', 'gogole')]
☐ [('listen', 'silent'), ('listen', 'enlist'), ('google', 'gogole')]
☐ [('listen', 'silent'), ('listen', 'enlist'), ('silent', 'enlist')]

No, the answer is incorrect.
Score: 0
Accepted Answers: [('listen', 'silent'), ('listen', 'enlist'), ('silent', 'enlist'), ('google', 'gogole')]

- 6) The following function is intended to check if two words are anagrams.

However, it fails for some valid inputs. Identify the bug.

```
def is_anagram(w1, w2):
    return sorted(w1) == sorted(w2)
```

What is the issue with this code?

- ☒ It doesn't handle uppercase and lowercase letters correctly
☐ sorted() can't be used on strings
☐ The return statement compares the memory addresses
☐ It requires the words to be of equal length

Yes, the answer is correct.
Score: 1
Accepted Answers: It doesn't handle uppercase and lowercase letters correctly

- 7) Which of the following must be true for two strings to be anagrams?

- ☐ Both strings must be in the dictionary
☒ Both strings must be the same length and have identical character counts
☐ Both strings must have the same starting letter
☐ Both strings must have the same number of vowels

Yes, the answer is correct.
Score: 1
Accepted Answers: Both strings must be the same length and have identical character counts

- 8) You are checking whether "abode" and "edcba" (note the space at the end) are anagrams. Which of the following best describes why a naive checker would fail?

- ☐ Because they have different lengths
☐ Because the space has ASCII value 32
☐ Because sorted lists don't match
☒ All of the above

Yes, the answer is correct.
Score: 1
Accepted Answers: All of the above

- 9) What will be the output of the following code?

```
def anagram_checker(word):
    return word == word[::-1]

print(anagram_checker("madam"))
```

- ☒ True
☐ False
☐ Error: invalid slicing syntax
☐ True, because "madam" is an anagram of "madam"

Yes, the answer is correct.
Score: 1
Accepted Answers: True

- 10) What will be the output of the following code?

```
def ascii_sum(word):
    total = 0
    for char in word:
        total += ord(char)
    return total

print(ascii_sum("abc") == ascii_sum("acb"))
```

- ☒ True
☐ False
☐ Error: ord() cannot be used in loop
☐ False, because the order of characters matters in ord()

Yes, the answer is correct.
Score: 1
Accepted Answers: True

A player randomly guesses a number from 1 to 10. The computer also picks a random number in the same range. If the guess is wrong, the score increases by 1. If it's correct, the score decreases by 1. This process repeats 10 times, and the score after each round is plotted on a graph.

```
import random
import matplotlib.pyplot as plt

counts = []
count = 0

for i in range(10):
    guess = random.randint(1, 10)
    pick = random.randint(1, 10)

    if guess != pick:
        count += 1
    else:
        count -= 1

    counts.append(count)

plt.plot(counts)
plt.show()
```

- 11) What is the range of numbers used for guessing in each round?

- ☐ 1 to 5
☐ 0 to 9
☒ 1 to 10
☐ 0 to 10

Yes, the answer is correct.
Score: 1
Accepted Answers: 1 to 10

- 12) What happens to the score if the player's guess matches the computer's pick?

- ☒ It increases by 1
☐ It stays the same
☐ It decreases by 1
☐ It becomes 0

No, the answer is incorrect.
Score: 0
Accepted Answers: It decreases by 1

- 13) How many values will be stored in the counts list after the loop finishes?

- ☐ 0
☐ 5
☐ 9
☒ 10

Yes, the answer is correct.
Score: 1
Accepted Answers: 10

- 14) Which of the following best describes the purpose of the final plt.plot(counts) line?

- ☐ It shows how many times the guess was correct
☐ It prints the final score
☒ It displays how the score changed over each round
☐ It compares the guess and pick values directly

Yes, the answer is correct.
Score: 1
Accepted Answers: It displays how the score changed over each round

- 15) Under which of the following conditions will the plot of the counts list form a straight line of the form $y = mx + c$ (either increasing or decreasing at a constant rate)?

- ☐ The guess matches the pick in some random rounds
☒ The guess never matches the pick throughout all 10 rounds
☐ The count increases or decreases based on a sine function
☐ The guess and pick values are both fixed at 5 in every round

Yes, the answer is correct.
Score: 1
Accepted Answers: The guess never matches the pick throughout all 10 rounds

- 16) Which of the following changes will make the score decrease with every guess?

- ☒ Replace count += 1 with count -= 1
☐ Make guess == pick always true
☐ Set count = 0 in every iteration
☐ Remove counts.append(count)

Yes, the answer is correct.
Score: 1
Accepted Answers: Replace count += 1 with count -= 1

- 17) What will happen if the line counts.append(count) is removed from the code?

- ☐ The graph will show the score progression
☐ The code will crash due to a missing list
☒ The counts list will remain empty, and the plot will be blank
☐ The program will enter an infinite loop

Yes, the answer is correct.
Score: 1
Accepted Answers: The counts list will remain empty, and the plot will be blank

- 18) What will happen if random.randint(1, 10) is replaced with random.randint(10, 1)?

- ☐ The guess will always be 10
☐ The range will be reversed and still work
☒ A ValueError will be raised due to invalid range
☐ The output will be a float instead of an int

Yes, the answer is correct.
Score: 1
Accepted Answers: A ValueError will be raised due to invalid range

- 19) How can you modify the code to simulate 100 guessing rounds instead of 10?

- ☒ Change range(10) to range(100)
☐ Replace random.randint(1, 10) with random.randint(1, 100)
☐ Increase the value of count by 100
☐ Run the code 10 times in a loop

Yes, the answer is correct.
Score: 1
Accepted Answers: Change range(10) to range(100)

- 20) How can you modify the code to plot both the guessed values and the picked values across rounds?

- ☐ Use a bar chart to show both values in one plot
☒ Store guess and pick values in separate lists and plot both using plt.plot()
☐ Plot only the counts list as it includes both guess and pick
☐ Multiply the guess and pick values and store them in one list

Yes, the answer is correct.
Score: 1
Accepted Answers: Store guess and pick values in separate lists and plot both using plt.plot()

Detective Neel was walking down the street when he noticed a blurry poster of a missing dog. The colors were faded, and the image barely recognizable. Curious, he decided to help. He took a photo of the poster with his phone and went back to his computer.

Neel had recently learned about a Python library called PIL—short for **Python Imaging Library**. It helped programmers **open**, **view**, **resize**, and **enhance** images using simple code.

First, he wrote:

```
from PIL import Image
```

This allowed him to access the Image class from the PIL module. He then opened the dog image with:

```
img = Image.open("dog_poster.jpg")
```

With just one line, he loaded the photo into Python! To make it easier to analyze, Neel resized it:

```
img_resized = img.resize((300, 300))
```

He printed out the format and mode of the image:

```
print(img.format) # e.g., JPEG
print(img.mode) # e.g., RGB or L
```

Finally, Neel displayed the image on screen using:

```
img_resized.show()
```

- 21) What does the PIL library allow users to do?

- ☐ Send emails
☒ Manipulate and view images
☐ Write games
☐ Draw graphs

Yes, the answer is correct.
Score: 1
Accepted Answers: Manipulate and view images

- 22) What was the purpose of Image.open("dog_poster.jpg")?

- ☐ To download an image
☐ To open the file in a browser
☒ To load the image into Python
☐ To delete the file

Yes, the answer is correct.
Score: 1
Accepted Answers: To load the image into Python

- 23) What does img.resize(300, 300) do?

- ☐ Crops the image
☐ Zooms into the image
☒ Resizes the image to 300x300 pixels
☐ Rotates the image

Yes, the answer is correct.
Score: 1
Accepted Answers: Resizes the image to 300x300 pixels

- 24) If img.mode prints "RGB", what does that mean?

- ☐ The image has only black and white shades
☒ The image uses Red, Green, and Blue channels
☐ The image is rotated
☐ The image is grayscale

Yes, the answer is correct.
Score: 1
Accepted Answers: The image uses Red, Green, and Blue channels

- 25) Which function is used to view the image on the screen?

- ☐ img.open()
☐ img.save()
☒ img.show()
☐ img.print()

Yes, the answer is correct.
Score: 1
Accepted Answers: img.show()

- 26) What will be the output type of this code?

```
from PIL import Image
img = Image.open("dog_poster.jpg")
print(type(img))
```

- ☐ < class 'str' >
☐ < class 'PIL._jpegImagePlugin.JpegImageFile' >
☐ < class 'Image' >
☒ < class 'PIL.Image.Image' >

No, the answer is incorrect.
Score: 0
Accepted Answers: < class 'PIL._jpegImagePlugin.JpegImageFile' >

- 27) What will the following code print?

```
img = Image.open("dog_poster.jpg")
img = img.resize((3, 3))
arr = np.array(img)
print(arr.shape)
```

- ☒ (3, 3)
☐ (3, 3, 3)
☐ 9
☐ Error: images cannot be converted to arrays

No, the answer is incorrect.
Score: 0
Accepted Answers: (3, 3, 3)

- 28) If the original image shape is (4, 4, 3), what would be the shape of the compressed array after slicing arr[:, :, 2]?

- ☒ (2, 2, 3)
☐ (2, 2)
☐ (4, 2, 3)
☐ (2, 4, 3)

Yes, the answer is correct.
Score: 1
Accepted Answers: (2, 2, 3)

- 29) What does each arr[x][y] contain in a colored image loaded with NumPy?

- ☐ A matrix
☐ A string
☒ A list of RGB values
☐ A single grayscale value

Yes, the answer is correct.
Score: 1
Accepted Answers: A list of RGB values

- 30) Which of the following is a correct sequence to open, resize, and view an image?

```
img.resize((100, 100))
Image.open("poster.jpg")
img.show()
```

```
img = Image.open("poster.jpg")
img = img.resize((100, 100))
img.show()
```

```
Image.open("poster.jpg").show()
img = resize(img, (100, 100))
```

```
open("poster.jpg")
resize(img)
show(img)
```

Yes, the answer is correct.
Score: 1
Accepted Answers:

```
img = Image.open("poster.jpg")
img = img.resize((100, 100))
img.show()
```